

Using Topological Data Analysis to Classify Pollen

Jay Lebakken, Ben Crotteau

Department of Mathematics, Statistics, and Computer Science
Macalester College
St. Paul, MN 55105

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Abstract

Pollen classification is the backbone of palynology, which is a field that can be applied in a broader environmental context to help research in areas such as climate change, evolution, biostratigraphy, forensics and paleoecology [1]. Currently, human experts are used to identify pollen species that have several drawbacks, including differences in interpretation due to subtle physical differences, long classification times, and the field losing collective knowledge upon retirement [1]. Automated methods for pollen identification have been proposed in recent years in response to the fact that a large percentage of experts are retiring [1]. The goal of these automated models is to increase the classification accuracy, increase the efficiency of the process, and preserve collective knowledge. We propose a topological approach to pollen identification in which pollen images are analyzed using a persistent homology technique called Vietoris-Rips filtration. Topology and persistent homology in particular are an effective tool in analyzing the shape of data that lend themselves to image analysis. The resulting structure identified by the Vietoris-Rips filtration was compiled in persistence diagrams corresponding to a pollen image. A measure of similarity between pairwise persistence diagrams was computed in order to classify the pollen images to a species group. Our model achieved 65.3 percent accuracy on 25 percent of the dataset. In addition to this, correctly classified species had a 53.3% lower bottleneck distance from other images of the same species than to other images of different species. Although other models and even human experts certainly have higher classification accuracies, we believe that this figure shows that persistent homology is effective at capturing the latent structure of pollen samples and should be further studied to be incorporated into future automated pollen identification methods.

1 Introduction

Palynological identification is important not only in the study of pollen, but also in a wide range of fields including biodiversity, paleoecology, climate change, evolution, forensics, and geology [1]. Pollen samples have several desirable traits that allow the study of these subjects. First, fossilized pollen samples extracted from rocks and sediments include hundreds of thousands of grains [1]. Second, correct classification of pollen species and morphology highlights environmental adaptations that provide ecological information for a given period of time [2].

However, palynological classification is difficult. Pollen taxonomic classifications have subtle morphological differences, and certain taxa have many similarities [1]. Currently, pollen identification is carried out manually by experts in the field [1]. The use of human experts to identify pollen species has several drawbacks. First, classifications are not always consistent from expert to expert [1]. Experts may have differing classifications due to their familiarity with the taxa, the impacts of

preservation on its physical appearance, and something as simple as their fatigue levels [1]. Second, this system limits the number of classifications to those that experts can reliably identify [3]. This impacts the scale and results of environmental analysis that can be conducted. Third, relying on human experts provides challenges in maintaining and transferring collective knowledge. When palynological experts retire, the field loses a source of knowledge.

Palynology is shifting from manual to automated pollen identification methods to address the need for increased consistency, accuracy, and a centralized container for collective knowledge. Several studies of automated pollen identification methods have shown potential promise in replacing human experts. Automated trait semantic modeling for image retrieval has been shown to be more consistent and accurate in pollen identification than human experts [[1]]. In this model, pollen images were analyzed to determine the shape and texture of the pollen grains [[1]]. Then an automated semantical model was used to determine the morphology of the pollen grain and annotate the image accordingly [[1]]. This allowed for image search from a database based on semantic similarity which would provide palynologists with a centralized pollen information source[[1]]. Importantly, these semantical methods had high accuracy rates and success in identifying subtle differences in images indicating the efficacy of image analysis for pollen identification [[1]].

Machine learning techniques such as convolutional neural networks (CNNs) have also been studied as a method for automated pollen identification. CNNs extract features from images in a hierarchical structure that encodes information about characteristics shape and texture. [[4]][[5]]. Furthermore, CNNs are effective in samples with high pollen species diversity and in samples with extra organic matter [6]. Similarly to trait semantic modeling, the CNN-based model was built on a database of annotated pollen images [6]. One potential drawback to the CNN model was that if it was trained on a dataset with highly uneven prevalence of different pollen taxa, its performance for a specific taxa will be tied to its representation in the original dataset [6]. Despite this drawback the use of CNNs significantly reduced classification time while increasing accuracy and consistency [4] [5].

Another potential method for pollen classification is topological data analysis, specifically persistent homology (PH). PH is a tool that analyzes the structure of data, and in the context of this paper we will be focusing on point clouds. There are several methods to computing persistent homology on point clouds, but we use Vietoris-Rips filtrations. At increasing steps of ϵ , the Vietoris-Rips complex is constructed from the k -simplices that correspond to unordered $(k + 1)$ -tuples of k of the points in the point cloud that are pairwise within distance ϵ [7]. Over all values of ϵ , the birth and death of n -dimensional features are monitored. These features are the n -cycles that are not n -boundaries. The features that persist through a relatively high number of filtration steps are interpreted to be more significant in highlighting the structure of a point cloud. These birth-death pairs can be compared to those from different point clouds to identify similarities and differences in the structure of data. A common application of persistent homology is image analysis, specifically identifying similarities and differences in images. The crux of the challenge of pollen identification is the ability to differentiate between subtle morphological differences in pollen and make accurate classifications based on that information. Thus, it seems that persistent homology could be another tool developed in palynological identification. It could potentially be used alongside machine learning tools, which will be addressed in the following sections. This study will address this question by evaluating the performance of persistent homology in pollen identification, and the suitability of persistence matrices as data for this task.

2 Methodology

2.1 POLLEN73S Dataset

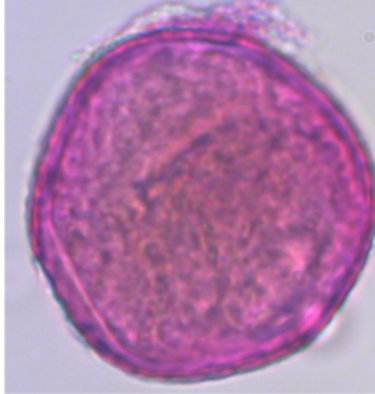
We use the POLLEN73S dataset to explore the homological structure of pollen images [8]. POLLEN73S contains 2523 low-resolution samples of 73 flowering pollen species from the Brazilian savanna [8]. The Brazilian savanna is one of the most biodiverse regions in the world which is reflected in the sample size of the dataset [8]. We chose this data set because it would be ideal for automated classification to be effective regardless of resolution. Low-resolution images for pollen samples are common because of large sample sizes and because electron microscope imaging may be expensive, unattainable, or unavailable.

2.2 Image Pre-Processing

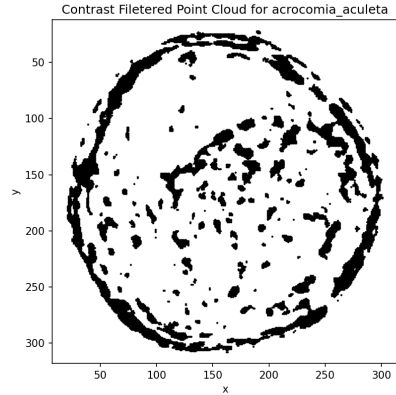
Pollen grain images in the POLLEN73S dataset have different backgrounds, capture angles, indicator colors, lighting, and contrast levels. To standardize these images, we first make the images black and white. Then, we create a point cloud PC from a pollen image P as a grid of coordinates (x, y) that fall outside one standard deviation of the mean light intensity using the following formula:

$$PC = \left\{ (x, y) | v(x, y) > \left(\frac{v(P)}{N} \right) + \sigma(P) \right\}$$

. Here, $v(x, y)$ is the given light intensity value at the point (x, y) , and σ represents the standard deviation of an image's light intensity value. The result is a "fingerprint" point cloud that captures the shape and contrast of a pollen sample. An example of this image processing for the species *acrocomia aculeata* is shown below in Figure 1. Pollen texture has been shown to be a distinguishing feature in pollen identification.



(a) Original Pollen Sample



(b) Generated Point Cloud

Figure 1: Point Cloud Creation from Pollen Image

2.3 Persistent Homology

After acquiring the "fingerprint" point cloud from the pollen sample images, we turned to persistent homology to analyze the structure of the point cloud which encoded information about the texture, shape, and contrast of the pollen samples. For each pollen sample, we computed the persistence diagram of features of dimension 0, 1, and 2. We passed the point clouds through the Vietoris-Rips filtration. From this filtration, 0, 1, and 2 dimensional features were born and died at varying steps in the filtration. Figure 2 below shows a visualization of the Vietoris-Rips filtration.

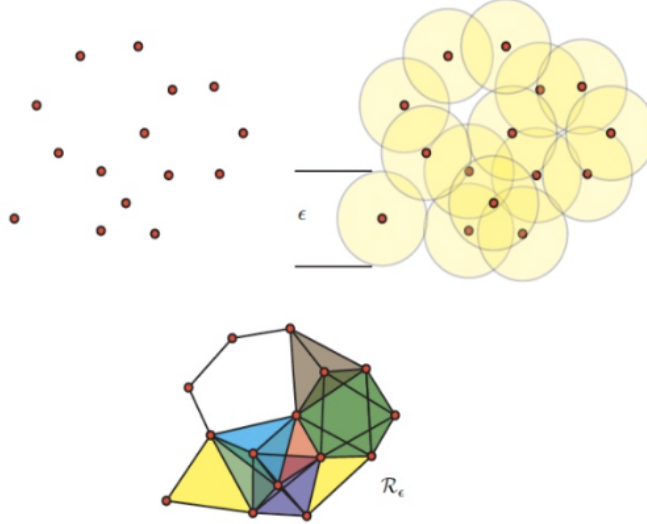


Figure 2: Vietoris-Rips Filtration Visualization [7]

To visualize this data, we used a persistence diagram that showed the death versus the birth for these 0, 1, and 2 dimensional features. Persistence diagrams are an effective method to visualize and analyze the results of data analysis using persistent homology because it conveys which features last longest in the filtration and thus are more indicative of the structure of the data.

Figure 3 below shows the persistence diagram for the species *acrocomia aculeata*. Analyzing the figure below, it is clear that nearly all 0-dimensional features, which are simply points in the point clouds, are born at the beginning of the filtration and die off in the early stages of the filtration. This makes sense as by the definition of Vietoris-Rips complexes, 0-dimensional features should instantaneously appear. Furthermore, 1- and 2- dimensional features often reside on the diagonal of the persistence diagram which signifies that they appear and disappear instantaneously. The 1- and 2- dimensional features that reside significantly above the diagonal represent features that persist under the filtration and are indicative of the structure of the point cloud. Our question is whether matching these persistence diagrams between samples will provide an effective and relatively simple method of pollen classification.

2.4 Pollen Classification

Bottleneck distance is the most straightforward method to compute the "distance" between persistence diagrams. Bottleneck distance is a "minimax" approach where points in two persistence diagrams are paired in order to minimize the largest distance between any two paired points. Despite being straightforward, bottleneck distance is computationally expensive to compute. We

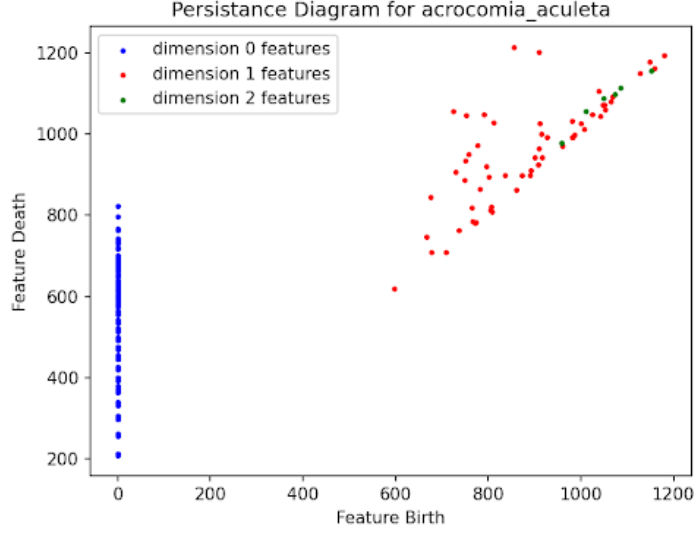


Figure 3: Persistence Diagram for a Pollen Sample

computed bottleneck distance between every pair of persistence diagrams, identifying the closest match. Figure 3 below shows a visual representation of this process.

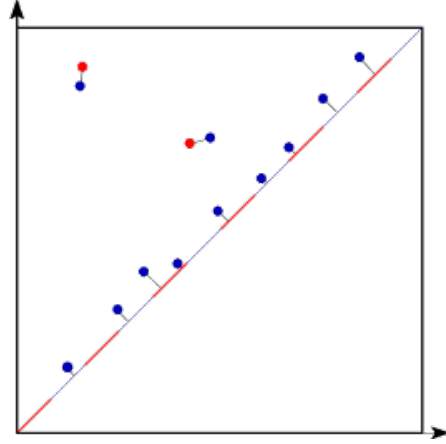


Figure 4: Bottleneck Distance Visualization [9]

3 Results and Discussion

Our model achieved 65.3% classification accuracy on a limited sample from our dataset. We only used 25% of our dataset to build the persistent homology model as computing raw bottleneck distance is computationally expensive and is $\mathcal{O}(n^4)$. Furthermore, this computation classified this 25% of the dataset in a few hours which is far less than the few days it takes human experts to complete the task.

There are several takeaways from our results. First, our persistent homology model appears to capture invariant features between pollen of the same species. The classification accuracy that we achieved was lower than that of other automated models and even human experts, but it was

also relatively simplistic and can be optimized as discussed below. The classification rate on this model suggests that persistence matrices of pollen data are a great candidate for machine learning models. [10]

Second, computing persistence matrices for the pollen data did not require manual feature selection like other machine learning or semantic trait modeling techniques [4] [1]. We were able to implement a blind model that was still able to capture information from pollen samples. Thus, if our model’s classification rates can be improved, it would be desirable for a model not to require any inputs such as manual feature selection.

Third, our model was implemented and tested on low-resolution pollen images such as those shown below in Figure 5. It is important for a model to be effective in low-resolution settings because high-resolution microscopy and imaging is often prohibitively expensive and inaccessible [4]. Thus, it is important for a model to be able to classify the large sample of low-resolution images that currently exists. Currently, research into automated methods for pollen identification has relied on high-resolution images [5]. As persistent homology appears to be effective method in a low-resolution setting, it could be incorporated into future automated methods to identify pollen.

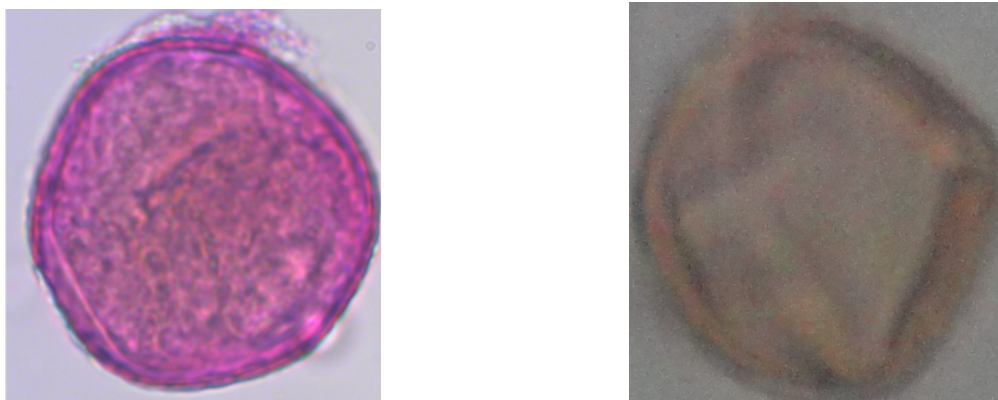


Figure 5: Example Images from POLLEN73S

4 Conclusions and Future Directions

From the results of our model, it appears that persistent homology is able to identify, track, and classify differing latent structural features in pollen samples. The accuracy that our model achieved was lower than that of other automated methods mentioned earlier, but there are several contextual pieces that suggest that persistent homology can be successfully used in a pollen identification setting. First, our model was relatively simplistic, and further optimization of the pipeline we introduced could increase both classification accuracy and speed. Our model didn’t explore other methods for computing persistent homology besides the Vietoris-Rips filtration. However, there are many options for filtrations, including building cubical simplices. These simplices are based on square shapes as opposed to triangles which is often applied to image data as pixels are square in shape.

Additionally, we only analyzed the persistence diagrams using bottleneck distance as well. Bottleneck distance is the most simplistic measure of ”distance” between two persistence diagrams, but there are machine learning techniques that have been developed to analyze persistence diagrams such as PerSlay. PerSlay is part of a group of methods that involve the vectorization of persistence diagrams for grouping and classification [10]. This vectorization is built on a DeepSets architecture

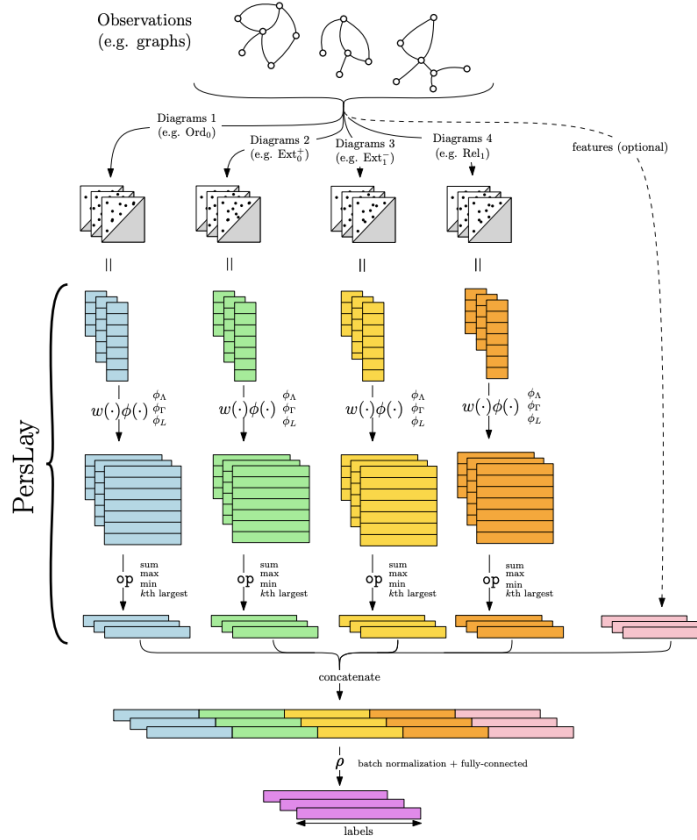


Figure 3: Network architecture illustrated in the case of our graph classification experiments (Section 4). Each graph is encoded as a set of persistence diagrams, then processed by an independent instance of PERSLAY. Each instance embeds diagrams in some vector space using two functions w, ϕ that are optimized during training and a fixed permutation-invariant operator op .

Figure 6: PerSlay Architecture Overview

to capture important information about its structure [10] [11]. Unsupervised machine learning techniques can be applied to these vectorizations for classification [10]. We believe that PerSlay will increase classification accuracy and efficiency in the palynology setting, so it is a candidate for future research to optimize our persistent homology pipeline. Below, Figure 6 provides an overview for PerSlay architecture.

Second, persistent homology appears to be effective for pollen classification on low-resolution images. The POLLEN73S dataset was comprised of low-resolution pollen images [8]. On the other hand, machine learning approaches to pollen classification have been studied on a dataset of high-resolution pollen images [4]. This could explain why the classification accuracy of our model was lower than that of the machine learning models. Furthermore, developing a method that can identify pollen at all resolution scales is important because high-resolution microscopy is expensive and often inaccessible. As a result, many datasets of pollen images are low-resolution. If researching and developing a more sophisticated persistent homology model could increase classification accuracy comparable to the machine learning models, it would be an advantage for that model to be able to classify low-resolution images, and it could potentially have even higher classification accuracies on high-resolution images.

Future research should be conducted to optimize the persistent homology pipeline. Persistent homology appears to be an effective method to classify pollen based on its physical structure in all resolution settings. Machine learning techniques for persistence diagrams such as PerSlay are one potential upgrade to our model as well as all the different filtrations that could be done instead of Vietoris-Rips such as cubical. There is a lot of promise in automating pollen identification as experts retire, and we believe that topology and persistent homology play an important role in the future of this field.

A Code

```
import os
from PIL import Image
import gudhi.bottleneck
import gudhi.hera
import matplotlib.pyplot as plt
import numpy as np
import gudhi
from gudhi.sklearn.rips_persistence import RipsPersistence as rp

def getCloud(image_ndArray):
    ##parameters, 2d Nd Array representing an image
    ##returns an array [x,y] representing a point cloud
    points = np.argwhere(image_ndArray<255)
    x = points[:, 1]
    y = points[:, 0]
    return [x,y]

def filter(path):
    #input: path of image
    #returns a filtered version of image
    img = Image.open(path)
    #Black and white conversion
    img = img.convert('L')
    imat = np.asarray(img)
    #filter image to points outside of standard deviation of image contrast
    imat = np.where(imat<(np.mean(imat)-np.std(imat)), imat,255)
    #Optional image saving:
    #img = Image.fromarray(imat,mode="L")
    #img.save("sampleImages/sampleimg_{}.png".format(int(i)))
    return imat

def showPoints(point_Cloud):
    #Parameter: [x,y] ndarray representing point cloud
    #Plots point cloud
    x = point_Cloud[0]
    y = point_Cloud[1]
    plt.figure(figsize=(6, 6))
```

```

plt.scatter(x, y, s=1, color='black')
plt.gca().invert_yaxis()
plt.axis('equal')
plt.title("Nonwhite - Pixels - Point - Cloud")
plt.xlabel("x")
plt.ylabel("y")
plt.show()

def computePD(filtered_image, dimensions=[0,1,2],
              input_type='point-cloud', homo_field=11):
    rips = rp(homology_dimensions=dimensions,
              input_type=input_type,
              num_collapses='auto',
              homology_coeff_field=homo_field, n_jobs=None)
    diag = rips.transform([filtered_image])
    return diag

def showPD(persistence_matrix, species):
    ##Param: persistence_matrix as 2d ndarray, species as string
    ##Plots persistence diagram
    ##unwrap PD
    PD = persistence_matrix[0]
    ##
    colors = ['BLUE', 'RED', 'GREEN']
    for i in range(0,3):
        xx = PD[i][:,0]
        yy = PD[i][:,1]
        plt.scatter(xx,yy, s=5, color=colors[i],
                   label=("dimension-" + str(i) + "-features"))
    plt.title("Persistence-Diagram-for-" + species)
    plt.xlabel("Feature-Birth")
    plt.ylabel("Feature-Death")
    plt.legend()
    plt.show()

def findMatch(image_path):
    #param: image path
    #returns: closest species match
    query_img = filter(image_path)
    #query_points = getCloud(query_img)
    #showPoints(query_points)
    query_PD = computePD(query_img)

    path = "Preprocessed_Data/"
    dataset = os.listdir(path)
    closesetmatch = ""
    min = 10000000000000000
    for folder in dataset:

```

```

bottleneck=0
average = 0
current_species = os.listdir(path+folder+"/")
for sample in current_species:
    dimension = 0
    for matrix in os.listdir(path+folder+"/"+sample):
        print(path+folder+"/"+sample+"/"+matrix)
        targetPD= np.load(path+folder+"/"+sample+"/"+matrix)
        bottleneck+= gudhi.hera.bottleneck_distance(targetPD,
            query_PD[0][dimension])
        dimension +=1
        average+=1
    print(bottleneck/average)
    if bottleneck/average<min:
        min=bottleneck/average
        closesetmatch = folder

return closesetmatch


def main():
    print(findMatch("Dataset/POLLEN73S/
-----anadenanthera_colubrina/anadenanthera_colubrina-(3).jpg"))


if __name__=="__main__":
    main()

```

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